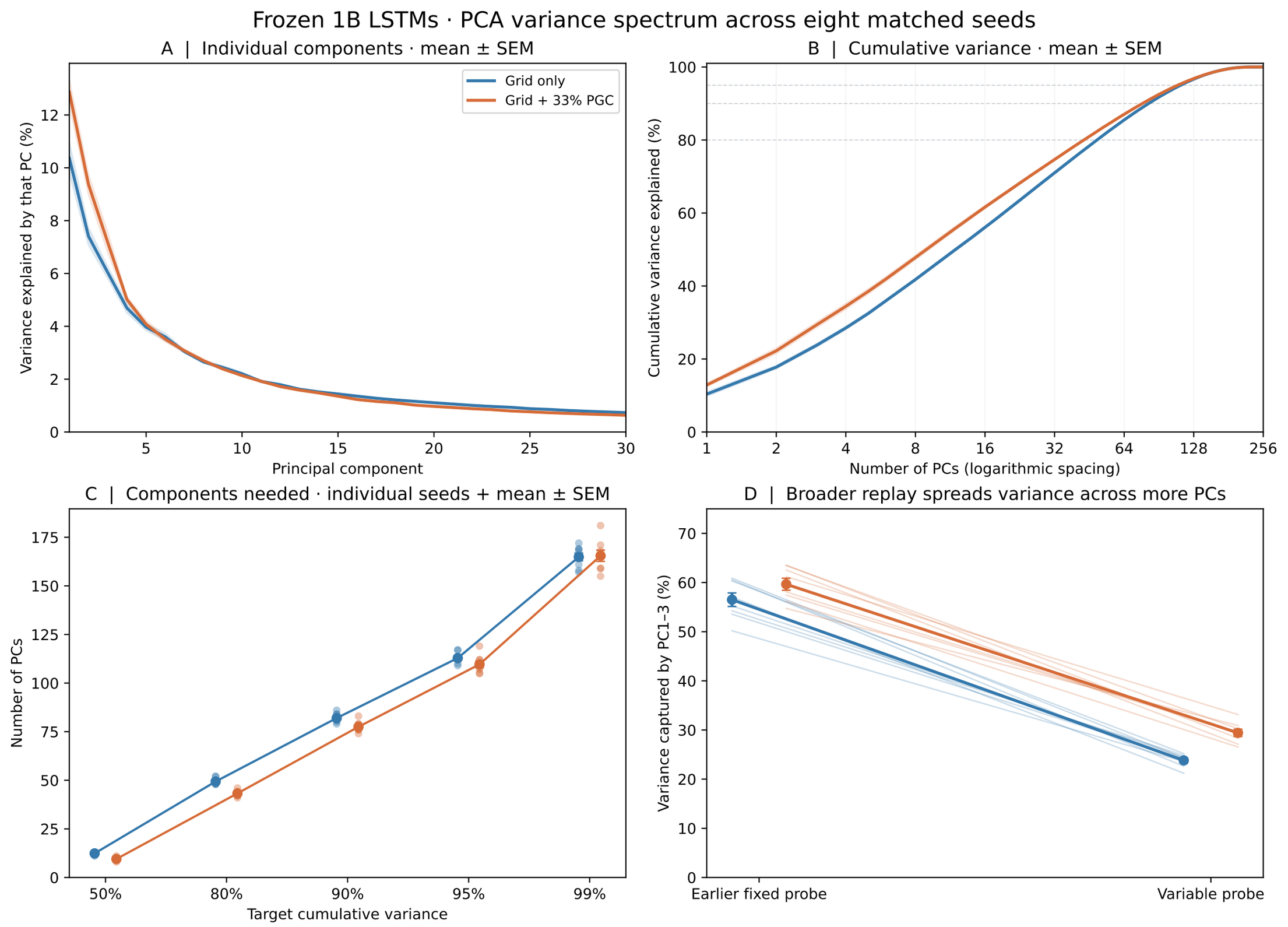
Frozen 1B LSTMs · PCA variance spectrum across eight matched seeds
A | Individual components · mean ± SEM
Grid only
Grid + 33% PGC
Variance explained by that PC (%)
Principal component
B | Cumulative variance · mean ± SEM
Cumulative variance explained (%)
Number of PCs (logarithmic spacing)
C | Components needed · individual seeds + mean ± SEM
Number of PCs
Target cumulative variance
50%
80%
90%
95%
99%
D | Broader replay spreads variance across more PCs
Variance captured by PC1–3 (%)
Earlier fixed probe
Variable probe
Centered h, no neuron scaling. PCA dimension measures linear variance, not gridness or torus dimension.
Fixed/variable probes also differ in trajectory, rooms, sampling, and episode structure.